PRIYANKA BOLEM

AI AND MACHINE LEARNING ENGINEER

LLM Applications | RAG | NLP | MLOps | Healthcare AI

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SUMMARY

AI and ML engineer with 6+ years in software engineering and AI/ML, shipping production LLM, NLP and predictive systems for healthcare, enterprise and banking clients. Built a real-time LLM assist that cut average handle time by about a third in a 2-million-call contact center. Strong in Python, SQL, PyTorch, RAG, LangChain and MLOps on AWS and Azure. MS in Computer Science.

TECHNICAL SKILLS

Languages: Python, SQL, Java, R

ML and Deep Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, Keras, Vector Databases

Modeling and Optimization: Classification, Regression, Clustering, Neural Networks, Reinforcement Learning, Bayesian Optimization, Monte Carlo Simulation, Uncertainty Estimation

Data and Pipelines: Apache Spark, Azure Databricks, Airflow, Kafka, Redis, ETL design, Feature Engineering, Microsoft Dynamics CRM

MLOps and Deployment: AWS SageMaker (Pipelines, Endpoints), MLflow, Docker, Kubernetes, FastAPI, GitHub Actions, CI/CD, Drift Detection

Cloud: AWS (SageMaker, Lambda, EC2, ECS, S3), Azure (ML Studio, Databricks), Microsoft Foundry, GCP (Vertex AI, BigQuery, Cloud Run)

Evaluation and Experimentation: A/B Testing, Cross-validation, ROC-AUC, Precision/Recall/F1, AUPRC, Model Calibration

Generative AI and NLP: LangChain, LangGraph, RAG, Prompt Engineering, Microsoft Foundry, Intent Classification, Summarization, BERT

Visualization: Tableau, Power BI, Matplotlib, Plotly, Seaborn, Pandas, NumPy, Statsmodels

PROFESSIONAL EXPERIENCE

AI Engineer | Premera Blue Cross (via Adisys) *Mountlake Terrace, WA | Jul 2026 - Present*

- Cut average handle time by about a third (13 to 15 minutes down to about 9) with a real-time LLM assist that summarizes the member's medical, vision and dental claims for the representative during the call.
- Designed the Redis data path between Genesys and the model that removes PHI and PII before the LLM.
- Built an intent classification model on member inquiries to route handling across voice, chat and email.
- Automated post-call documentation, the summary and inquiry record representatives once wrote by hand.
- Building the member portal chatbot and IVR voicebot on the Genesys APIs, driven by the intent model (Portkey, Microsoft Foundry, Redis, Dynamics CRM, AWS).

Machine Learning Engineer | HP Inc. (via Adisys) *Remote | Dec 2025 - Jun 2026*

- Built the sales and service application for Fiji, HP's internal sales and service system, connecting its data through APIs to Microsoft Foundry for AI-driven decisions over HP's business rules.
- Embedded LinkedIn data into Dataverse with AI recommendations, removing manual account lookups so sales representatives close deals faster.
- Analyzed millions of Fiji project and factory records with AI and ML to find lifecycle improvements.
- Built dashboards presenting the AI analytics on sales and operations to leadership.

Machine Learning Engineer | Truveta (via Adisys) *Seattle, WA | Feb 2025 - Nov 2025*

- Deployed end-to-end ML pipelines on AWS SageMaker (Pipelines for training, Endpoints for real-time risk scoring, CloudWatch monitoring) to healthcare data compliance standards.
- Built Python and TensorFlow models forecasting chronic disease progression from de-identified EHR data, automating patient stratification for clinical research.

- Improved plan efficiency and outcome prediction accuracy by 22% with a hybrid framework pairing Genetic Algorithms for cohort selection with Reinforcement Learning.
- Cut model training and data preparation time by 35% by optimizing SQL and pipelines on a cloud data lake.

Machine Learning Engineer | Capgemini *India | Apr 2019 - Nov 2022*

- Cut loan decision cycle time by 25% with Scikit-learn, XGBoost and LightGBM models for credit default, delinquency and fraud, embedded in origination, collections and fraud triage systems.
- Cut data preparation time by 40% with automated Python and SQL ETL across core banking, credit bureau and CRM systems; scaled distributed training on Azure Databricks.
- Reduced false positives by 20% with a Python monitoring system tracking accuracy, drift and alert thresholds across credit and fraud models.
- Orchestrated Apache Airflow workflows for scheduled retraining, data refresh and batch inference.

Junior Software Engineer | Halcyon Technologies *India | Feb 2018 - Mar 2019*

- Built and maintained Java Android apps, implementing features and fixing defects under senior review.

PROJECTS

- **Financial Sentiment MLOps Platform.** End-to-end NLP platform: FinBERT fine-tuning, MLflow tracking, FastAPI + Docker + Kubernetes serving, Prometheus/Grafana monitoring, Evidently drift detection, GitHub Actions CI/CD; live on Hugging Face Spaces. github.com/priyankabolem/financial-sentiment-mlops • live demo
- **Grocery Product Search and Re-ranking on Google Cloud.** Two-stage search over an Instacart catalog in BigQuery: Vertex AI Search retrieval, a gradient boosting re-ranker on a Vertex AI endpoint scoring reorder propensity, and a FastAPI service on Cloud Run blending both into the final ranking with a retrieval-order fallback; Docker, tests, CI. github.com/priyankabolem/grocery-search-vertex
- **Credit Card Fraud Detection: Production MLOps.** Fraud scoring on a 1:578-imbalanced dataset, chronological split: XGBoost vs logistic baseline, test AUPRC 0.765 vs 0.718 and ROC-AUC 0.979, threshold set for a 0.90 precision budget, FastAPI serving, PSI drift monitoring, Docker, CI with 10 tests. github.com/priyankabolem/fraud-detection-mlops

EDUCATION

Master of Science, Computer Science | Northwest Missouri State University, Maryville, MO *Jan 2023 - Dec 2024*